

Theory of Automata

CS-3005

Lecture 8

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FAST NUCES CFD

Epsilon NFA

- ϵ -NFAs add a convenient feature but (in a sense) they **bring us nothing new**. They do not extend the class of languages that can be represented.
- Both **NFAs and E-NFAs recognize** exactly the same languages.

Convert this in NFA

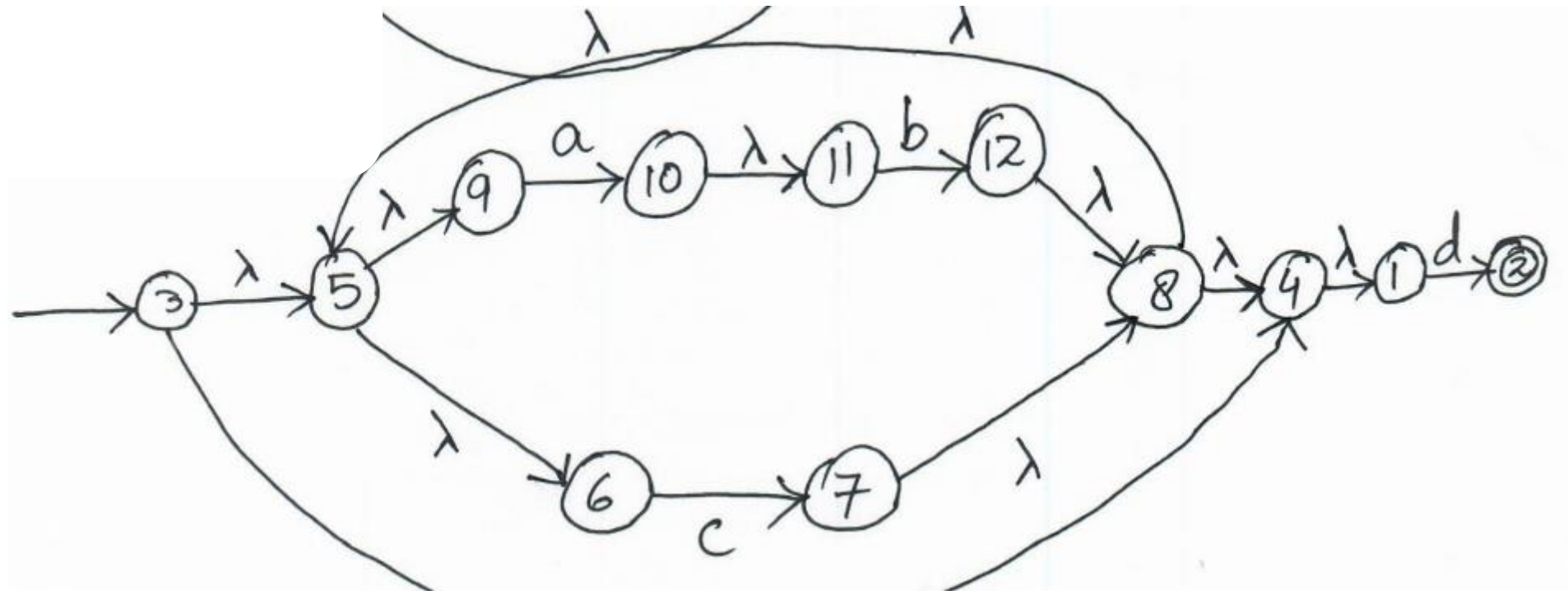
- Regular Expression

$(ab + c)^*d$

Respected NFA:



Convert the RE in Epsilon NFA



Transition Table

States	a	b	c	d	Λ^*
1				2	1
2					2
3					3,5,6,9,4,1
4					4, 1
5					5, 6, 9
6			7		6
7					7,8,4,1,5,6,9
8					8,4,1,5,6,9
9	10				9
10					10, 11
11		12			11
12					12, 8, 4, 1, 5, 6, 9

Simplify Transition Table

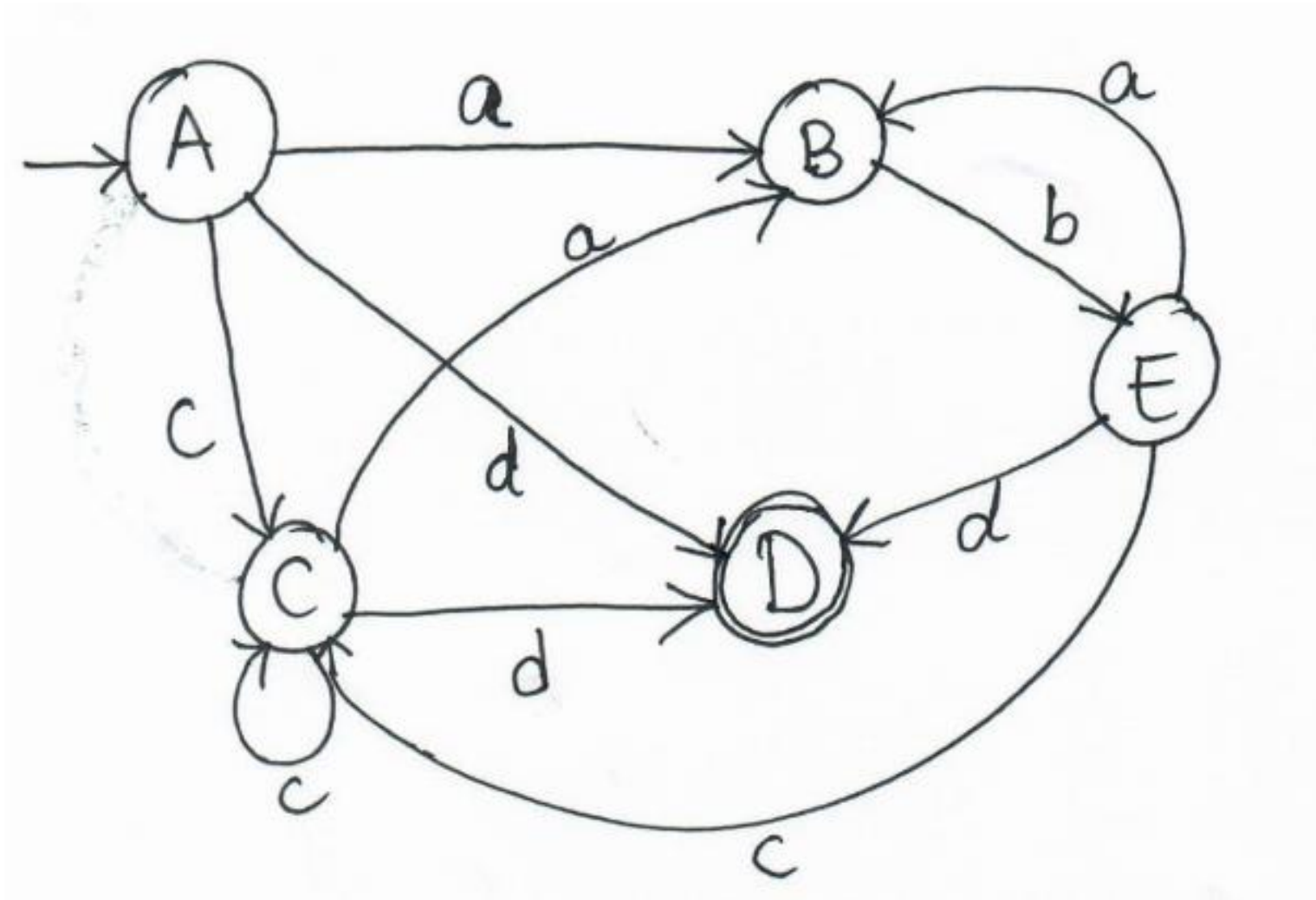
States	a Λ^*	b Λ^*	c Λ^*	d Λ^*
3,5,6,9,4,1	10, 11		7,8,4,1,5,6,9	2
10, 11		12, 8, 4, 1,5,6,9		
7,8,4,1,5,6,9	10, 11		7,8,4,1,5,6,9	2
2				
12, 8, 4, 1,5,6,9	10, 11		7,8,4,1,5,6,9	2

States	States name
3,5,6,9,4,1	A
10, 11	B
7,8,4,1,5,6,9	C
2	D
12, 8, 4, 1,5,6,9	E

Updated Transition Table

States	$a\Lambda^*$	$b\Lambda^*$	$c\Lambda^*$	$d\Lambda^*$
A	B		C	D
B		E		
C	B		C	D
D				
E	B		C	D

DFA



All else go to the Garbage state

State Minimization of DFA

- DFA minimization stands for converting a given DFA to its equivalent DFA with minimum number of states. DFA minimization is also called as Optimization of DFA and uses partitioning algorithm.

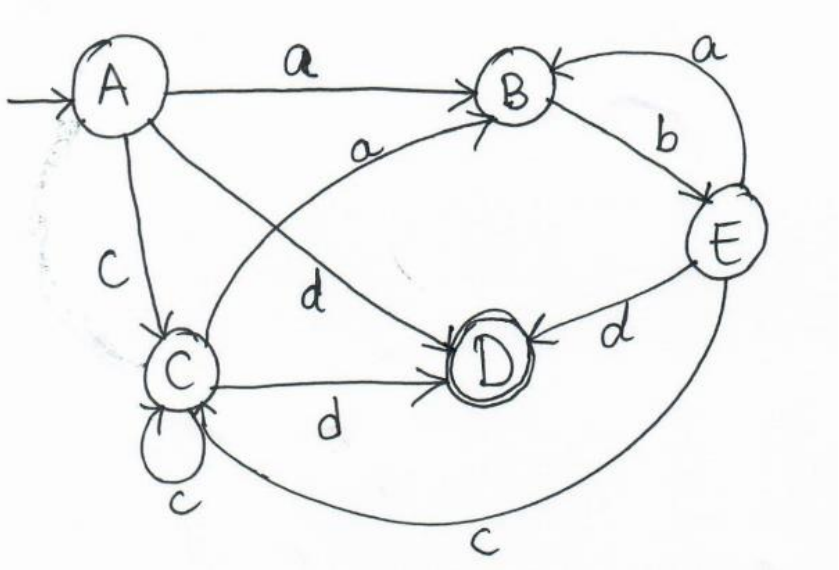
State Minimization of DFA

P			
A	B	C	E
B-CD	-E--	B-CD	B-CD
P-PQ	-P--	P-PQ	P-PQ

Q
D

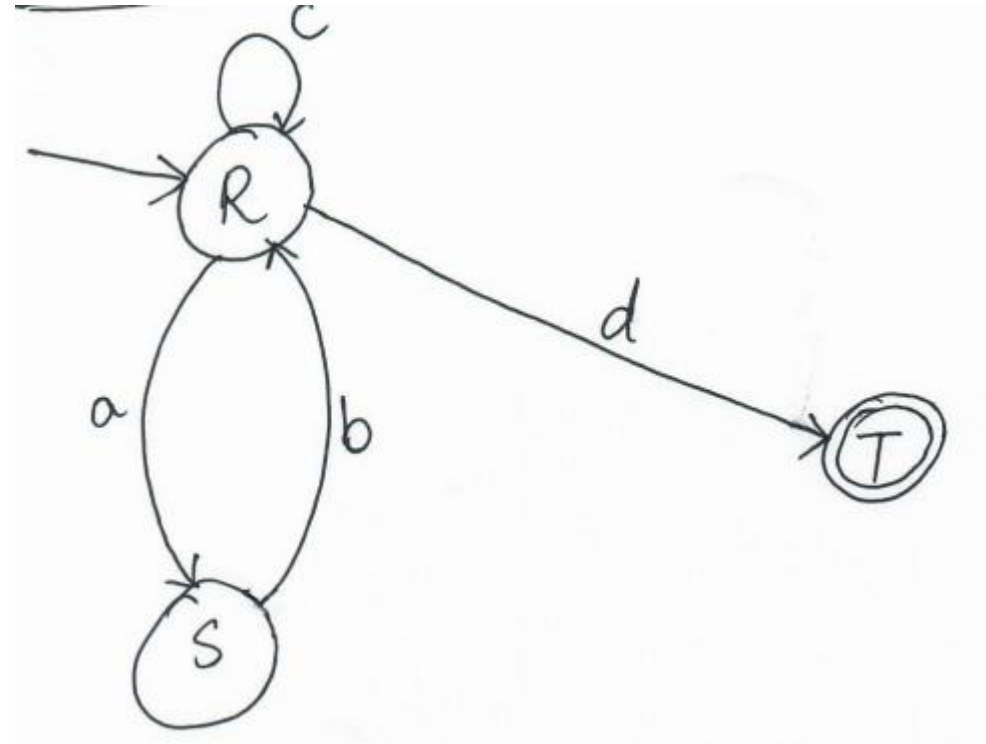
R		
A	C	E
B-CD	B-CD	B-CD
S-RT	S-RT	S-RT

S	T
B	D
-E--	----
-R--	



State Minimization of DFA

	A	B	C	E
R	S		R	T
S		R		
T				



All else go to the Garbage state